Python Assignment
Topic-9: String Manipulation

Q1. Understanding how to access and manipulate strings.

Ans.

Strings are used to store text data in Python. We can access characters using index numbers starting from 0. String manipulation means changing or using strings in different ways. We can convert text to uppercase or lowercase. We can also join or replace text. Strings cannot be changed directly, so new strings are created. String manipulation is useful in programs where text is used. It is commonly used in input and output operations.

Example:

```
text = "Anshu"
print(text[0])
print(text.upper())
```

Q2. Basic operations: concatenation, repetition, string methods.

Ans.

Python provides many operations for working with strings. Concatenation is used to join two strings using the plus operator. Repetition is done using the star operator to repeat a string. String methods like upper(), lower(), and replace() are used to modify strings. These operations make string handling easy and flexible. They are very useful in text processing and formatting.

Example:

```
a = "Hello "
b = "Anshu"
print(a + b)
print(b * 2)
print(a.upper())
```

Q3. String slicing.

Ans.

String slicing is used to get a part of a string. It is done using index values inside square brackets. We can give starting and ending positions to extract characters. It is very useful when working with text data. If we do not give index values, Python uses default values. Slicing makes string operations easy and powerful. It is an important concept in Python programming.

Example:

```
name = "Anshu"
print(name[1:4])
print(name[:3])
print(name[-1])
```